

Lost in Dristor 2 (dristor2)

This is the legend of the old, long-lost package in Dristor 2.

A long time ago, AAP, MH, MFast and the other package hunters were asked to find a lost package in the Dristor 2 station. There are N package hunters and M secret locations in the Dristor 2 station, each package hunters know how to reach some secret locations. Since none of them know the station well enough, they decided to split up to increase their chances of finding the package.



Figure 1: Dristor 2 station.

In a search, every package hunter check at most one secret location known. Every hunter must check a different secret location and the number of secret locations checked must be the maximum possible.

Count the number of different searches the group can try in order to find the lost package. Since this number can be very large, you should print the remainder of the division by $10^9 + 7$. Two searches are considered different if at least one package hunter checks a different location.

 Among the attachments of this task you may find a template file `dristor2.*` with a sample incomplete implementation.

Input

The first line contains the two integer N and M .

The next N line contains M integers $L_{i,0}, L_{i,1}, \dots, L_{i,M-1}$. $L_{i,j}$ is 1 iff the hunter i knows the secret location j , 0 otherwise.

Output

You need to write a single line with an integer: the number of different searches modulo $10^9 + 7$.

🔒 The *modulo* operation ($a \bmod m$) can be written in C/C++/Python as `(a % m)` and in Pascal as `(a mod m)`. To avoid the *integer overflow* error, remember to reduce all partial results through the modulus, and not just the final result!
Notice that if $x < 10^9 + 7$, then $2x$ fits into a C/C++ `int` and Pascal `longint`.

Constraints

- $1 \leq N \leq 14$.
- $1 \leq M \leq 100$.
- $0 \leq L_{i,j} \leq 1$ for each $i = 0 \dots N - 1$ and $j = 0 \dots M - 1$.

Scoring

Your program will be tested against several test cases grouped in subtasks. In order to obtain the score of a subtask, your program needs to correctly solve all of its test cases.

- **Subtask 1** (0 points) Examples.

- **Subtask 2** (35 points) $N, M \leq 6$.

- **Subtask 3** (35 points) Each hunter know every secret location.

- **Subtask 4** (30 points) No additional limitations.


Examples

input	output
4 5 0 1 0 1 0 1 1 0 0 1 1 0 0 0 0 0 1 0 1 1	3

Explanation

In the **first sample case**, the possible searches are:

- $0 \rightarrow 1, 1 \rightarrow 4, 2 \rightarrow 0, 3 \rightarrow 3$
- $0 \rightarrow 3, 1 \rightarrow 4, 2 \rightarrow 0, 3 \rightarrow 1$
- $0 \rightarrow 3, 1 \rightarrow 1, 2 \rightarrow 0, 3 \rightarrow 4$